

ALEXANDER KIRCHBERGER

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EDUCATION

Western University

May 2023 - August 2025

M.E.Sc in Electrical and Computer Engineering

London, ON

- Performed research into efficient macromodeling techniques of linear electric systems, specifically when systems are electrically long or characterized by noisy tabulated frequency responses.
- Publications include: Instrumental Variable Vector Fitting Method for Rational Approximation of Noisy Data in IEEE Microwave and Wireless Technology Letters and Delay Rational Macromodeling of Noisy Tabulated Frequency Responses at EPEPS 2024 conference.
- Instructed and marked course assessments, such as laboratories project reports and exams as a graduate teaching assistant for ECE 2277 Digital Logic Systems and ECE 3375 Microcontrollers.

Western University

September 2018 - April 2023

B.E.Sc in Electrical Engineering - with Co-op and Distinction

London, ON

WORK AND ENGINEERING EXPERIENCE

Western University - Western Formula Racing

May 2023 - April 2025

Grounded Low Voltage Lead

London, ON

- Led the design assembly and validation of grounded low voltage system, and supported design and assembly of the high voltage system of an Formula SAE car.
- Designed and validated system architectures, PCBs, wiring harnesses, and high voltage components.
- Work was performed under intense time and resource constraints.

Western University - Western Engineering Build Team

September 2022 - April 2025

Electrical Team

London, ON

- Designed and assembled professional light shows for prop structures under intense time and resource constraints.
- Designed, troubleshoot, and repaired legacy systems for orientation week.

General Dynamics Land Systems - Canada

May 2021 - August 2022

Electrical Design Specialist (Internship)

London, ON

- Design lead of 3 subsystems of a prototype light armored vehicle.
- Designed 138 wiring harnesses and cables, circuit protection, grounding schemes, and component selection.
- Reviewed networking architectures for issues related to hardware compatibility.
- Performed EMI testing for MIL-STD-461 at the device and vehicle level.

Western University

May 2020 - August 2020

Undergraduate Summer Research Internship

London, ON

- Developed a high impedance fault detection scheme using Mathematical Morphology.

PROJECTS

WFR-24E ECU

- Main controller for the Western Formula Racing Formula SAE car, based on a Teensy 4.1 development board.
- The controller interfaces with all the components within the car, such as the motor inverter over CAN.
- For this year the car was converted from a one controller system to a 2 controller system to reduce overall wiring harness complexity and specifically the wiring from front to rear.

WLED Light Controller

- ESP32 based light controller, utilizing open source WLED software used for light shows.
- The controller contains 4 powered pixel outputs capable of controlling up to 4048 pixels.

TECHNICAL SKILLS

Circuit Design

HSPICE, LTSPICE, Altium, KiCAD, PCB Design, RF Design, Signal Integrity Design

Model Generation

Vector Fitting, Transient Simulation, Electromagnetic Simulation

Software

C, Python, MATLAB, Rust, Lua, Bash, RTOS, Linux, Vim, Git